

Question	Answer	Marks
3(a)(i)	signal consists of (a series of) 1s and 0s or offs and ons or highs and lows	B1
3(a)(ii)	component X: parallel-to-serial converter	B1
	component Y: DAC/digital-to-analogue converter	B1
3(a)(iii)	sample the (analogue) signal	M1
	at regular intervals and converts the analogue number to a digital number	A1
3(b)(i)	attenuation in fibre = 84×0.19 (= 16 dB)	C1
	ratio = $16 + 28$ = 44 dB	A1
3(b)(ii)	ratio / dB = $10 \lg (P_2 / P_1)$	C1
	$44 = 10 \lg (\{9.7 \times 10^{-3}\} / P)$ or $-44 = 10 \lg (P / \{9.7 \times 10^{-3}\})$	C1
	power = $3.9 \times 10^{-7} \text{ W}$	A1

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4(c)	random/horizontal	D1